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Beyond Learned Embeddings: Evaluating Sinusoidal Encodings for Parameter-Efficient Byte-Level Recurrent Networks

Final Model Performance and Expressiveness

The central inquiry into the feasibility of replacing learned embeddings with fixed sinusoidal encodings in a byte-level recurrent neural network hinges on whether the model can maintain or achieve parity in final performance metrics such as loss and accuracy. The analysis reveals that this transition is highly probable, supported by strong theoretical arguments and compelling empirical evidence from related architectures, particularly transformers and character-level LSTMs. While a direct study on the specified ring-buffer pointer network is not available in the provided sources, the principles derived from analogous systems provide a robust foundation for prediction. The core mechanism enabling this potential success is the ability of the downstream network components to effectively interpret and utilize the structured information provided by the fixed encoding, thereby compensating for the absence of a trainable lookup table.

A primary source of confidence stems from the widespread and successful application of fixed sinusoidal positional encodings in the Transformer architecture [18] [233]. Although originally designed to inject positional information, their fundamental principle—providing a deterministic, frequency-based signal to represent discrete entities—is directly relevant to the proposed use case of representing token values [2] [142]. The original Transformer paper established that these fixed encodings could enable the model to generalize to sequence lengths not seen during training, demonstrating their capacity to carry meaningful structural information without being learned from scratch [195][198]. This success suggests that a rich, mathematical structure like sine and cosine waves can serve as an effective representation, provided the subsequent layers are capable of extracting useful features from it. The key insight is that the informational content and its organization, rather than its learnability, determine its utility for the model. For a byte-level vocabulary of 0-255, the task of the embedding is less about capturing complex semantic nuances and more about providing a unique, high-dimensional vector for each of the 256 possible values. Sinusoidal encodings are well-suited for this, generating a

diverse set of vectors based on different frequencies that inherently capture some notion of similarity between nearby integers [115].

Direct empirical validation comes from a study on Long Short-Term Memory (LSTM) language models, which provides the most relevant evidence for recurrent networks [199]. In this research, an LSTM was tasked with a reverse-ordering problem, requiring it to reverse a sequence of integers. A critical finding was that an LSTM equipped with sinusoidal positional encoding successfully reversed sequences of length 64, even with a vocabulary size as large as 16,384, achieving over 95% token-wise accuracy. Conversely, a vanilla LSTM without positional encoding failed as the vocabulary size increased [199]. This experiment is highly instructive because it demonstrates that an RNN can leverage a fixed, non-positional signal (in this context, applied to token values) to learn a complex, abstract rule. It empirically validates the hypothesis that the model can "learn around" a fixed input representation by relying on its internal state and transformation layers to derive meaning. This finding strongly supports the expectation that a recurrent network can compensate for the fixed nature of sinusoidal encodings. The user's own intuition—that the first layer can learn any rotation to map the fixed signal into a useful space—is corroborated by the literature. The authors of the aforementioned study suggest the effectiveness of positional encoding arises from its "autonomous nature," being invariant to input values, which helps the network discover general patterns rather than memorizing specific instances [199]. In the proposed architecture, where the input encoding feeds directly into a `tanh`-based hidden state update (`hidden = tanh(input + ring_read + old_hidden)`), this initial transformation becomes paramount for interpreting the incoming sinusoidal vector.

Despite this optimism, it is crucial to acknowledge the theoretical concept of expressiveness loss associated with fixed representations [7]. Learned embeddings possess the flexibility to adapt to the specific statistical properties and semantic relationships within the training data, creating an optimized representation space. Sinusoidal encodings, by contrast, offer a generic, pre-defined structure. However, for the specific challenge of a 256-token byte-level vocabulary, the magnitude of this potential loss is likely minimal. The primary function required is disambiguation—a unique vector for each byte value. Sinusoidal encodings excel at this, producing 8192-dimensional vectors that are distinct and decorrelated enough to serve this purpose effectively. The "loss" is not necessarily a catastrophic failure if the downstream recurrent model is powerful enough to interpret and exploit the inherent structure of this fixed signal. Furthermore, the compensation mechanism, where the first few layers of the network learn to project the sinusoidal input into a manifold suitable for the recurrent core, appears to be a viable strategy. This aligns with the observation that in some cases, the first layer of a neural

network can effectively learn any necessary rotation or transformation of the input space [194]. Therefore, while a theoretical expressiveness gap exists, practical evidence from related tasks suggests it may be bridged by the model's architecture and optimization process.

Impact on Overfitting and Generalization

Replacing a substantial portion of a model's parameters—4.2 million out of 5.3 million in the specified case—with fixed encodings represents a significant architectural change with profound implications for model complexity and, consequently, its tendency to overfit. The core hypothesis posits that this reduction in free parameters will act as a powerful regularizer, narrowing the performance gap between the training and evaluation sets. This conclusion is grounded in fundamental principles of machine learning, where model capacity is inversely related to generalization performance on unseen data [207]. However, the practical outcome is nuanced and depends heavily on the interplay between the model's reduced capacity and the inherent simplicity of the fixed encoding scheme. The literature indicates that overfitting was a major concern for early neural networks, especially those with tens of thousands of parameters trained on relatively small datasets, underscoring the importance of controlling model complexity [105].

The primary mechanism through which fixed encodings mitigate overfitting is the direct reduction in the number of trainable weights. With fewer parameters, the model has a diminished capacity to memorize idiosyncrasies of the training data, forcing it to focus on learning more generalized, underlying patterns [229]. In the scenario described, removing the two largest embedding-related layers (Input Embedding and Output Projection) constitutes a ~79% reduction in the total model parameters [239]. This drastic simplification should theoretically constrain the hypothesis space the model explores during training, making it less prone to fitting noise. The effect is akin to applying techniques like dropout or weight decay, which also limit the model's degrees of freedom to improve generalization [198]. For datasets that are sufficiently large and diverse, this regularization effect is expected to yield a clear benefit: a smaller train/eval loss gap and improved performance on held-out data. The saved parameters would otherwise contribute to the model's capacity to fit random fluctuations in the training set, so eliminating them directly reduces this risk.

However, this benefit is not unconditional and carries a potential trade-off towards underfitting. If the fixed sinusoidal encodings are too simplistic to adequately represent the necessary information for the task, the model may struggle to learn even the basic patterns in the data, leading to poor performance on both the training and evaluation sets. The success of this strategy therefore depends on a delicate balance. On one hand, the parameter reduction helps prevent overfitting. On the other hand, if the fixed encodings lack sufficient expressive power for the specific byte-level problem, the model may fail to converge to a good solution at all. The size and complexity of the dataset play a critical role here. For very large-scale problems where overfitting is the dominant challenge, the regularization benefit of parameter pruning is likely to dominate. For smaller or noisier datasets, the risk of underfitting due to insufficient representational capacity becomes more pronounced. The literature notes that early deep learning models often suffered from overfitting precisely because they had too many parameters relative to the amount of available data [105], suggesting that modern models often operate in a regime where reducing parameters is beneficial.

Another subtle aspect to consider is the role of the embedding layer in the training dynamics. Learned embeddings introduce a form of adaptive feature engineering, and the gradients flowing through this layer can add a degree of stochasticity to the optimization process. This noise can sometimes help the model escape sharp, poorly generalizing minima. By removing this layer, the optimization landscape might become smoother, potentially altering how the model navigates towards a final solution [187]. While the primary effect is expected to be a reduction in overfitting, this change in dynamics could have secondary consequences on the quality of the final model. Ultimately, the net effect on overfitting is a probabilistic outcome. Based on established regularization theory, a significant parameter reduction should reduce overfitting. The empirical validation of this hypothesis requires a controlled experiment comparing the train and validation loss curves of the learned-embedding and fixed-encoding models. The plot of these curves would visually demonstrate whether the gap narrows as expected. If the fixed-encoding model achieves similar or better final performance with a consistently smaller gap, it would provide strong evidence for the effectiveness of this technique as a regularizer in the target architecture.

Influence on Convergence Dynamics

The investigation into the impact of fixed sinusoidal encodings on convergence speed reveals a compelling and consistent trend across multiple studies: the use of fixed

encodings accelerates the training process compared to learned embeddings. This finding stands in contrast to the intuitive assumption that removing a trainable component might slow down learning. Instead, the evidence points towards a more efficient optimization trajectory, driven by the simplification of the model's objective function and the stability of the input signal. While the most direct evidence comes from transformer-based models, the underlying mechanisms are general enough to be highly relevant to the proposed recurrent architecture.

A large-scale comparative study on Automatic Speech Recognition (ASR) provides the most direct evidence for this phenomenon [198]. The study found that Transformer models, which employ fixed sinusoidal positional encodings, exhibited significantly faster training iteration rates—up to eight times faster on a single GPU—compared to RNN baselines. The authors attribute this speedup to several factors, including the elimination of a trainable layer, which simplifies the gradient flow and reduces computational overhead per step. More importantly, the fixed nature of the encoding provides a stable, pre-computed signal to the network. Unlike learned embeddings, whose values fluctuate dynamically during training, the sinusoidal encoding remains constant. This stability allows the remaining trainable parameters to converge more rapidly toward an optimal configuration without having to co-adapt with a changing input representation. This observation is critical because it isolates the effect of the encoding method itself, holding the overall model architecture largely constant.

The mechanism behind this accelerated convergence is straightforward. When embeddings are learned, the optimizer must simultaneously adjust the embedding matrix and all other parts of the network. The embedding layer acts as a foundational component; its gradients must propagate correctly to ensure that the entire system learns a coherent representation. This can create a complex and potentially unstable optimization landscape. By fixing the encoding, this entire sub-network is removed from the trainable set, simplifying the overall optimization problem. The model's task becomes solely focused on learning the transformations from this fixed input signal to the desired output. This principle is a common theme in model compression and efficiency techniques, where removing trainable components often leads to faster training [187]. The cleaner, more consistent gradient signals flowing into the first layers of the network can lead to more stable updates and a quicker descent towards a good local minimum.

While the ASR study compared Transformers and RNNs, the core finding regarding the benefit of a fixed input signal is likely to transfer to the proposed ring-buffer recurrent network. The key distinction between the architectures—the presence of self-attention in the Transformer—is less relevant to the specific mechanism of convergence acceleration related to the input encoding. The benefits stem from the static nature of the signal itself,

not the subsequent processing mechanism. In the user's architecture, where the sinusoidal vector feeds directly into a `tanh`-activated hidden state, the advantages are preserved. The first transformation layer receives a stable, predictable input, allowing it to stabilize quickly. This can manifest in several ways: reaching a target loss threshold in fewer epochs, requiring a lower learning rate to maintain stability, or exhibiting less noisy loss curves during training. The literature highlights that training stability and learning rate choice are critical for convergence, and a simpler optimization problem generally makes this process easier [91] [174]. Therefore, it is highly probable that the proposed change will result in a measurable improvement in convergence speed, offering a significant advantage beyond just parameter reduction. This represents a crucial practical benefit that should be measured empirically alongside the other key metrics.

Comparative Analysis of Alternative Encoding Strategies

To fully evaluate the efficacy of fixed sinusoidal encodings, it is essential to benchmark them against alternative input representation strategies, namely the current "one-hot + linear projection" baseline and the combined approach of using fixed encodings with weight tying. This comparative analysis illuminates the specific contributions of the sinusoidal structure and helps isolate the effects of parameter reduction from the benefits of a particular encoding type. The provided materials allow for a detailed examination of these alternatives, revealing distinct trade-offs in terms of representational power, training dynamics, and parameter efficiency.

The "one-hot + linear projection" approach serves as the direct equivalent of the currently used learned embedding layer. One-hot encoding transforms each token into a high-dimensional, sparse vector where only a single element is active [17] [210]. When this sparse vector is multiplied by a learnable linear projection matrix, the result is a simple lookup operation that retrieves a row from the matrix, identical to a standard learned embedding table [19] [173]. Therefore, this baseline is functionally equivalent to the current model setup and represents the most appropriate point of comparison. The primary difference between this baseline and the proposed fixed sinusoidal encoding lies in the nature of the initial representation. One-hot vectors are orthogonal and disconnected; there is no inherent relationship encoded between different token indices. In contrast, sinusoidal encodings produce dense, smooth vectors where nearby integers receive similar but not identical representations [63]. This geometric property is hypothesized to be a key reason for the faster convergence observed in transformer models [198]. A direct empirical comparison would measure the cost of this smoothness: does the slightly more

complex initialization of sinusoidal encodings lead to superior final performance, or is the primary benefit limited to faster training dynamics? Given that the byte-level vocabulary is small, the expressiveness requirements are low, and the main functional need is simply to provide a unique identifier for each byte. Both methods can satisfy this requirement, but the sinusoidal method adds an extra layer of implicit structure that may or may not be beneficial for the specific task.

Weight tying, a common practice in transformer-based language models, offers another avenue for parameter reduction [64]. This technique involves sharing the same weight matrix between the input embedding layer and the final output linear classification layer [227]. This halves the parameters dedicated to input/output mappings and has been shown to sometimes improve convergence [64]. The question arises whether this technique is effective with a very small vocabulary of 256 tokens. The literature suggests that separating input and output embeddings can be beneficial, as the model may learn specialized representations for encoding the input context versus decoding the output probability distribution [124]. However, this benefit is likely to be marginal for a small vocabulary. With only 256 classes, the output projection matrix is relatively small, and the model is unlikely to gain significant advantage from specializing the representations. The shared matrix would still be capable of learning an effective mapping in both directions. Therefore, applying weight tying in the context of a byte-level model is a plausible strategy that would further reduce the parameter count, complementing the savings from fixed encodings. Combining fixed sinusoidal inputs with a tied output projection presents an interesting hybrid approach. Here, the input is mapped from a fixed table, and the logits are generated by projecting the hidden state onto this same fixed table. This would result in zero learnable parameters for the entire input/output pipeline, maximizing parameter efficiency. The success of this combined approach would depend on whether the fixed sinusoidal table is rich enough to serve dual purposes: as an informative input representation and as a discriminative basis for the output classifier. The empirical evidence from LSTMs shows that a fixed encoding can support complex tasks, lending some support to this ambitious goal [199].

The following table summarizes the characteristics of the different encoding strategies under consideration:

Strategy	Input Representation	Learnable Parameters (I/O)	Key Advantages	Key Disadvantages
Baseline (One-Hot + Linear)	One-hot vector passed through a learnable linear layer [19].	4.2M (Input Emb. + Output Lin.)	Standard approach; maximum flexibility for the model to learn any representation.	High parameter count (~79% of model); slower convergence potential.
Fixed Sinusoidal Input	Deterministic 8192-dim vector from sin/cos functions [18].	2.1M (Output Linear only)	Significant parameter reduction; potential for faster convergence [198]; stable input signal.	Potential for expressive power loss; requires the model to compensate via other layers.
Fixed + Weight Tied	Deterministic 8192-dim vector from sin/cos functions.	0M (Shared I/O table/projection)	Maximum parameter reduction; maximizes capacity of core model.	Highest risk of expressive power loss; relies entirely on fixed table for both input and output.

This comparison clarifies the strategic choices. The move from the baseline to fixed sinusoidal input is a trade-off between maximum flexibility and efficiency. The subsequent step to include weight tying is a further push towards maximal efficiency, carrying an increased risk but also the highest potential reward in terms of freeing up model capacity. Empirical testing of these three configurations side-by-side would provide definitive answers to their relative merits in the context of the specific recurrent network.

Synthesis and Strategic Recommendations

This comprehensive analysis of replacing learned embeddings with fixed sinusoidal encodings in a byte-level recurrent network synthesizes findings from transformer and recurrent network literature to address the core research questions of final performance, overfitting behavior, and convergence speed. The collective evidence strongly suggests that this architectural modification is not only feasible but also holds tangible benefits, primarily in convergence speed and parameter efficiency, with a high likelihood of maintaining competitive final performance. The primary uncertainty stems from the lack of direct experimental studies on this exact combination of byte-level data and a recurrent ring-buffer architecture, necessitating cautious extrapolation of findings from related domains. The architectural differences, specifically the absence of self-attention and the direct feeding of inputs into a `tanh`-activated hidden state, mean that while core principles are expected to hold, the magnitude of the effects remains to be empirically verified.

Regarding final performance, the outlook is positive. The success of fixed sinusoidal positional encodings in state-of-the-art Transformer models demonstrates the viability of

using structured, non-learned signals to convey critical information to a neural network [18] [233]. More directly, an experiment on LSTMs showed that a recurrent network could successfully learn a complex abstract rule (sequence reversal) when provided with a fixed sinusoidal signal, proving its ability to compensate for a non-adaptive input representation [199]. This validates the "compensation" hypothesis, where the model's trainable layers learn to map the fixed encoding into a useful space. For the specific challenge of a 256-byte vocabulary, the expressiveness loss of sinusoidal over learned embeddings is expected to be minimal, as the primary task is disambiguation, a function for which sinusoidal encodings are well-suited.

On the question of overfitting, the analysis confirms that the dramatic reduction in parameters (~79%) will act as a powerful regularizer, which is expected to narrow the train/eval performance gap [207][229]. This is a classic regularization effect, limiting the model's capacity to memorize the training data. The practical outcome, however, depends on a balance: for large datasets where overfitting is the primary concern, the benefit is clear. For smaller or more complex datasets, there is a risk that the fixed encodings might be too simplistic, potentially leading to underfitting. An empirical comparison of the train and validation loss curves is essential to quantify this effect in the target domain.

The most confident prediction pertains to convergence speed. Evidence from a large-scale ASR study indicates that models using fixed sinusoidal encodings exhibit significantly faster training iteration rates compared to their learned-embedding counterparts [198]. This acceleration is attributed to the simplified optimization landscape created by removing a trainable layer and providing a stable input signal. This mechanism is general and is highly likely to apply to the proposed recurrent architecture, promising a notable improvement in training efficiency beyond mere parameter savings.

Based on this synthesis, the following strategic recommendations are proposed for conducting the empirical investigation:

1. **Design a Controlled Experiment:** Create a controlled experiment to directly compare four distinct configurations:

- **Baseline:** The current model using learned input and output embeddings.
- **Sinusoidal Input:** Replace the input embedding with fixed sinusoidal encodings while keeping the output projection learned.
- **One-Hot Baseline:** Implement the "one-hot + linear projection" approach to serve as a direct functional equivalent of the baseline.

- **Combined (Sinusoidal + Weight Tied):** Use fixed sinusoidal inputs and tie the output projection weights to the fixed encoding table (or a separate linear layer).

1. **Track Priority Metrics Rigorously:** During training, meticulously monitor and record the three key metrics in order of priority:

- **Final Performance:** Track the validation loss and/or accuracy until the model converges to ensure parity with the baseline.
- **Overfitting Behavior:** Plot the training and validation loss curves for each configuration to quantitatively measure the train/eval gap.
- **Convergence Speed:** Record the number of epochs and/or wall-clock time required for each model to reach a predefined performance threshold.

1. **Acknowledge Extrapolation:** In reporting the findings, clearly document that supporting evidence is drawn from transformer and standard RNN literature. Explicitly frame the conclusions as indicative or highly probable, pending the new experimental results, and highlight the architectural differences (recurrence vs. attention) as a key contextual factor.

By executing this plan, the research can move from a well-grounded theoretical prediction to a set of concrete, actionable findings that definitively answer the user's research goal, providing clear guidance on the feasibility and practical benefits of adopting fixed sinusoidal encodings in the specified recurrent network.

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